



RAFFLES INSTITUTION
H2 Computing (9569)
2025 Year 5

Tutorial S3B: Relational Database

1. A hospital keeps data on the patients, staff and wards.
 - (a) Identify 3 entities in the database, and 3 or more attributes for each entity.
 - (b) Write the table definitions using standard notation, identifying the primary key of each table.
 - (c) Draw an E-R diagram to show the relationships between the entities.
2. Information is to be held by a car-hire firm about cars and all the long-term hirers that a particular vehicle has had. A sample of the data in its unnormalised state is as follows:

CUSTOMER

Customer Number	Customer Name	Company Name	Town	Postcode	Car Registration	Make	Model	Date Hired	Date Returned
153	Nazarali,N	MF Plastics	Derby	DB53ER	J986NDX	Ford	Mondeo	12/2/21	25/8/21
					K788NRT	Ford	Mondeo	1/3/21	1/4/22
187	Brown,L	MF Plastics	Derby	DB53ER	L346NDS	Volvo	440	2/4/22	
287	Pinner,M	Tooke Ltd	Derby	DB68YU	J986NDX	Ford	Mondeo	15/9/21	14/8/22
					M411NGT	Renault	Espace	15/9/21	1/2/22

The structure for this table may be written as

CUSTOMER (Customer Number, Customer Name, Company Name, Town, Postcode, [Car Registration, Make, Model, Date Hired, Date Returned])

Fields within brackets [...] indicate repeating fields.

- (a) Write the table as two separate entities in first normal form.
 - (b) Create a 'link' table and write the three entities in 2nd normal form.
 - (c) Create a fourth entity to put the data in third normal form.
3. A company selling stationery keeps its orders on a database. An order may consist of several lines, as shown below.

order-number	date	product-code	description	qty-ordered	unit-price
152117	23-5-21	DG8766	eDictionary	1	222.95
		SG1121	speakers	2	120.00
		CD1127	crayons	3	9.99

- (a) Identify the entities involved and draw an entity-relationship diagram.
 - (b) Show using standard notation the tables in the database after normalisation.
4. A company keeps the following details on its stock items and suppliers. Some stock items may be obtained from any one of a number of suppliers.
 - Stock code
 - Description
 - Price
 - (and for each supplier:) Supplier code
 - Name
 - Address
- (a) Draw a diagram showing the relationship between stock item and supplier.
 - (b) Show how this information would be held on a database in third normal form.

5. The details about staff, students, courses and rooms in a college are as follows:

- Students may take a number of courses
- Each course has only one member of staff teaching that course
- Staff can teach more than one course
- Each member of staff has their own teaching room which is not used by any other member of staff
- The date that a student enrolls on a course is recorded

Four tables can be identified: Student, Course, Course List and Staff.

- Draw the E-R diagram and explain how these four tables are linked.
- Write the table descriptions of the four tables in standard notation.
- Explain how the details of all the students taught by a named member of staff can be found.

6. The details about chefs, customers, dishes and tables in a restaurant are as follows:

- Customers may order a number of dishes
- Each dish has only one chef cooking that dish
- A chef can cook more than one dish
- Each chef has their own table which is not used by any other chef to serve their dishes
- The date that a customer orders a dish is recorded

Four relational tables in a database can be identified: Customer, Dish, Dish-Orders and Chef.

- Specify the attributes (fields) required and state the primary key for each table.
- Draw the E-R diagram and explain how the tables are linked.
- Explain how the details of all the customers served by a named chef can be found.

7. Each year students enrol on many college courses. The data is stored in a database which has not been normalised; some of the data is inconsistent. An example of the data is shown.

Student Number	Name	Address	Tel Num	DOB	Course Title
101	Ian Smith	22 Apple Drive	91234567	29/07/92	H2 Physics
101	Ian Smith	22 Apple Drive	91234567	29/07/92	H2 Chemistry
123	Jane Toh	31 Farm Road	91112222	17/03/92	H2 Mathematics
101	Ian Smyth	22 Apple Drive	91234567	29/07/92	H2 Computing
123	Jane Toh	31 Farm Road	91112222	17/03/92	H2 Physics
136	Harry Lim	44 Tree Street	92322334	22/05/92	H2 Biology
123	Jane Toh	31 Farm Road	91112222	17/03/92	H2 Computing
.....					

- Explain data inconsistency, using an example from the above table.
- Explain data redundancy, using an example from the above table.
- Explain the difference between a primary key and a foreign key in a relational database.

8. A real estate agency sells apartments. This agency stores data in a relational database. The agency also stores a Customer table and an Apartment table. The Apartment table contains the details of apartments that are available for sale and the Customer table contains details of people who want to buy apartments.

(a) Specify the attributes required for the Customer and Apartment tables.

A customer may visit many apartments, and each apartment may be visited by many customers.

(b) Show the relationship between Customer and Apartment.

(c) Draw an E-R diagram between these tables that would satisfy the requirements of a relational database. Show the relations in standard notation.

(d) The agency would also like to maintain a list of apartments that have been sold and keep track of the customers who have bought these apartments. Explain how the E-R diagram and relations above can be modified to accommodate this.

9. [9754/2008/02/06]

The data about a sports club is kept in a database. The club runs a number of TEAMS.

Each TEAM has a number of MEMBERS playing for it and one or more COACHes.

A MEMBER can only play for one TEAM but a COACH may train more than one TEAM.

(a) Draw a fully labelled ER diagram to show how the entities TEAM, MEMBER, COACH are related, while keeping data redundancy to a minimum.

(b) Using examples taken from this application explain what is meant by:

- (i) a primary key or key attribute;
- (ii) a foreign key;
- (iii) a composite key.

10. [PJC/Prelim/9597/2016/02/06]

The following table shows some data that are stored in a college.

Student no	Student Name	Programme	Programme Duration (years)	Module No	Module Name	Lecturer
13828	Elvin Gan	P302	2	M165 M121	Visual Arts Networking	Fang Jason
13253	Goh Seng Lee	P305	2	M121 M110	Networking Database	Jason Kabu
13423	Yong Kee Le	P502	3	M181 M107 M110	Music Accounting Database	Sunny Honto Kabu
13098	Mahesh Babu	P306	4	M121 M110	Networking Database	Jason Kabu

A student enrolled in a programme may take several modules.

A module is only delivered by one lecturer.

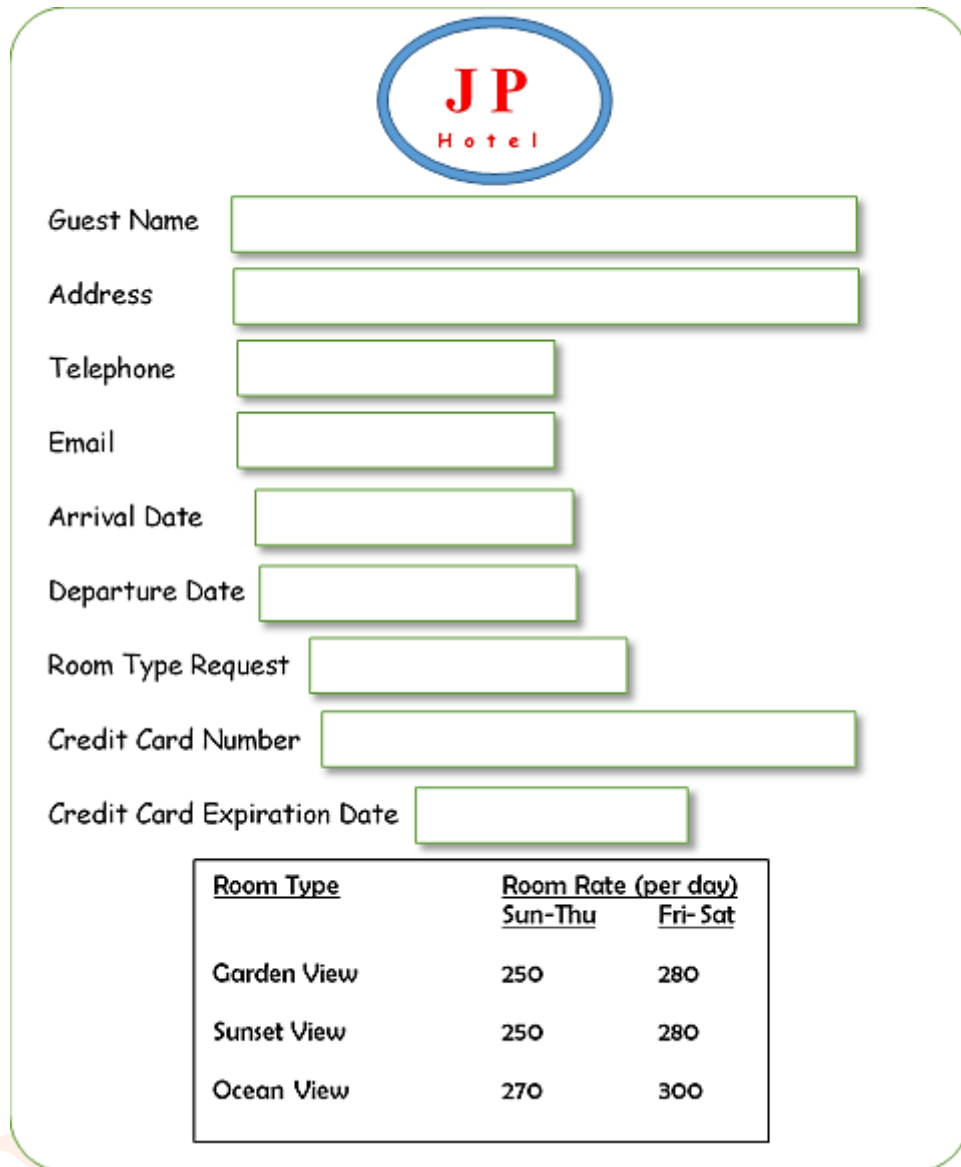
(a) The data in the table are in its un-normalised form. Explain three problems associated with data redundancy.

(b) Normalise the database and write them in four tables.

(c) Draw an ER diagram that shows the relationships between these four tables.

11. [JPJC/Prelim/9597/2019/02/06]

A reservation form used for booking JP Hotel rooms is shown:



The reservation form for JP Hotel includes the following fields:

- Guest Name
- Address
- Telephone
- Email
- Arrival Date
- Departure Date
- Room Type Request
- Credit Card Number
- Credit Card Expiration Date

Below the form is a table showing room types and rates:

Room Type	Room Rate (per day)	
	Sun-Thu	Fri-Sat
Garden View	250	280
Sunset View	250	280
Ocean View	270	300

There are three room types available, and the room rates are higher for arrival on Fridays and Saturdays.

A normalised database solution is designed to store data for the hotel using several tables.

- (a) Draw an E-R diagram that shows these tables and the relationships between them.
 (b) Using standard notation, write the table descriptions of the tables in part (a).

12. [PJC/Prelim/9597/2018/02/09]

PJ Mall plans to create a database to store data on its shops. It rents out shops to tenants who run their business.

- Each *tenant* is to provide information on its *company name*, *director of company*, *company address*, *contact number*, and *retail type*.
- There is a *start* and *end date* for every rental.
- Each *shop* rented by the tenant consists of one or more unit spaces.
- Each unit space is located at a particular *level* and has a *unit number*.
- There are 3 categories of unit space. Each *category* has its own *size* and *rental rate*.

Category	Size (square feet)	Rental rate (\$ per square feet)
A	Less than 200	40
B	200 – 2000	30
C	More than 2000	20

Here are some tenants who run their business in PJ mall:

Company Name	Level	Unit Number	Retail Type
Bata	2	03 – 04	Footwear
Challenger	2	06 – 08	Technology
Coldwear	3	08	Fashion
Esprit	3	09 – 10	Fashion
Giant	1	01 – 12	Supermarket
Hi Tea	1	14	Food & Beverage
PappaRich	2	11 – 13	Food & Beverage
... ..	...	...	...

(a) A solution is to create a relational database which requires a number of tables to store data for this application.

(i) Draw the E-R diagram showing the tables and the relationships between them.

(ii) A table description can be described as

TableName (Attribute1, Attribute2, Attribute3,)

The primary key is indicated by underlining one or more attributes.

Write table descriptions for the tables in part (i).

(b) Describe two advantages of using a relational database for storing data on its shops rather than a customised software.